

Class05: Data Viz with ggplot2

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Background

There are many graphic systems in R for making plots and figures These include so-called “*base R*” *graphics* like the `plot()` function and on packages like **ggplot2**.

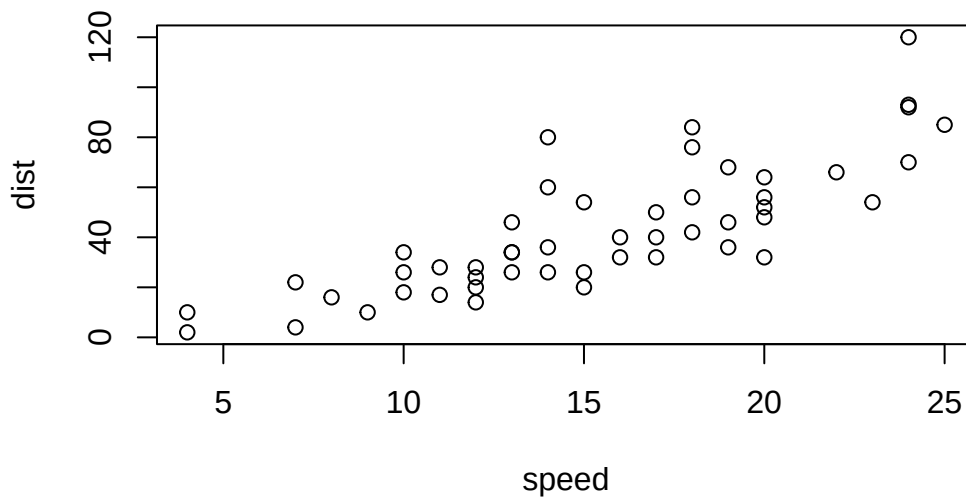
Let’s compare how we make a simple figure with these two systems:

We can use the in-built `cars` dataset:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

```
plot(cars)
```

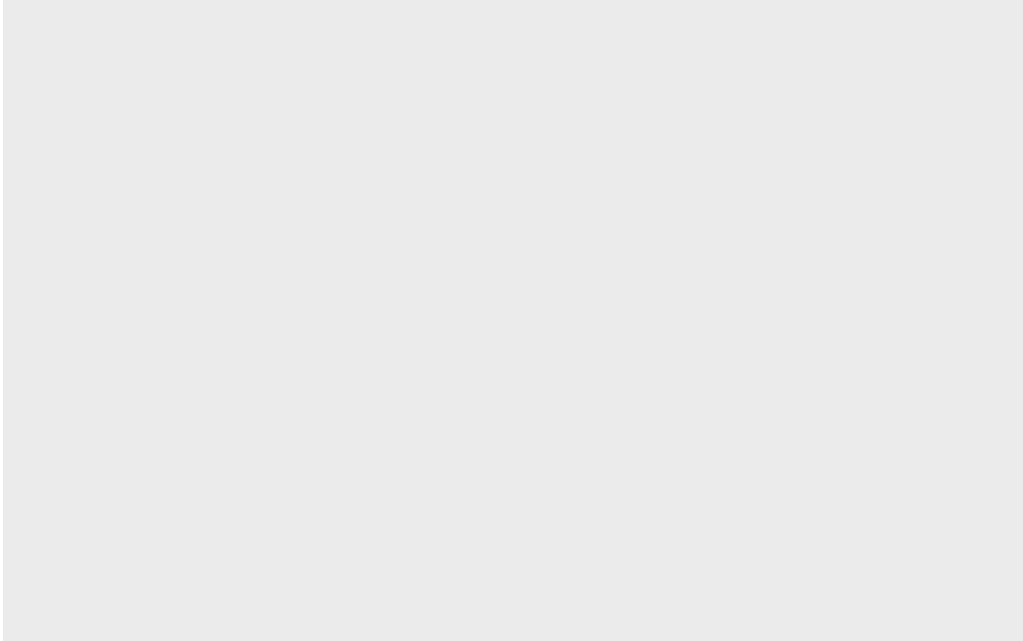


Before I can use ggplot I need to install it on my computer. To do this we can use the function `install.packages("ggplot2")` >**N.B** We never run `instal.packages()` in our quarto doc we run it once only in our R console as it would re-install the package every time we render out quarto report once installed we need to load up the package into our R brain:

```
library(ggplot2)
```

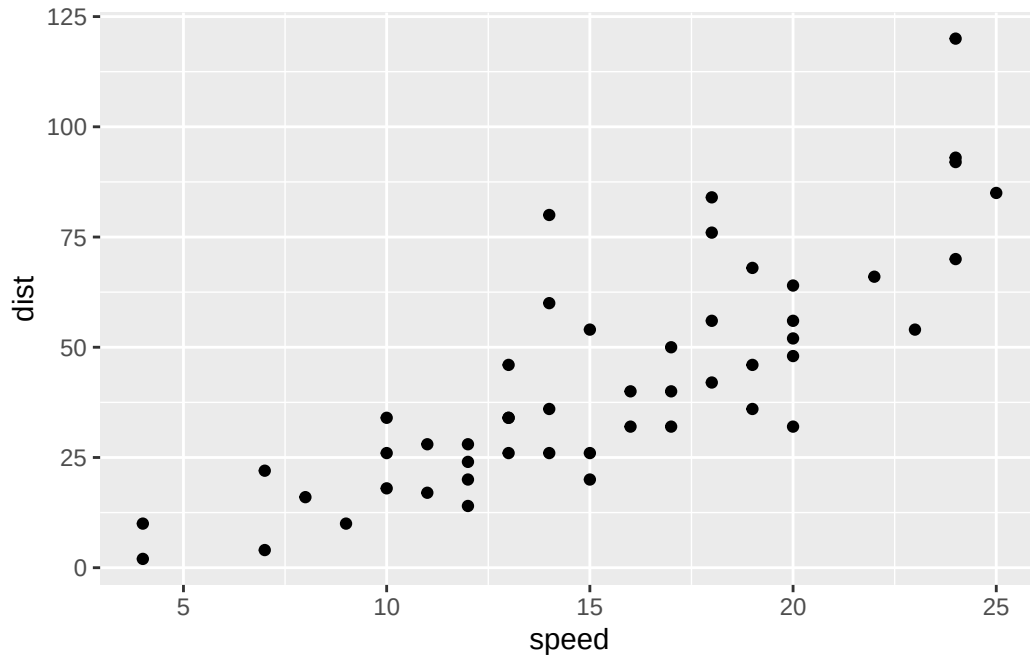
The main function in the **ggplot2** package is called 'ggplot()'

```
ggplot(cars)
```



Every ggplot has at least 3 layers: the **data** (a data.frame of the stuff we want to plot) the **aes** thetics(how the data maps to the plot), the **geom** layer (how you want the plot drawn, e.g. points, lines, etc.)

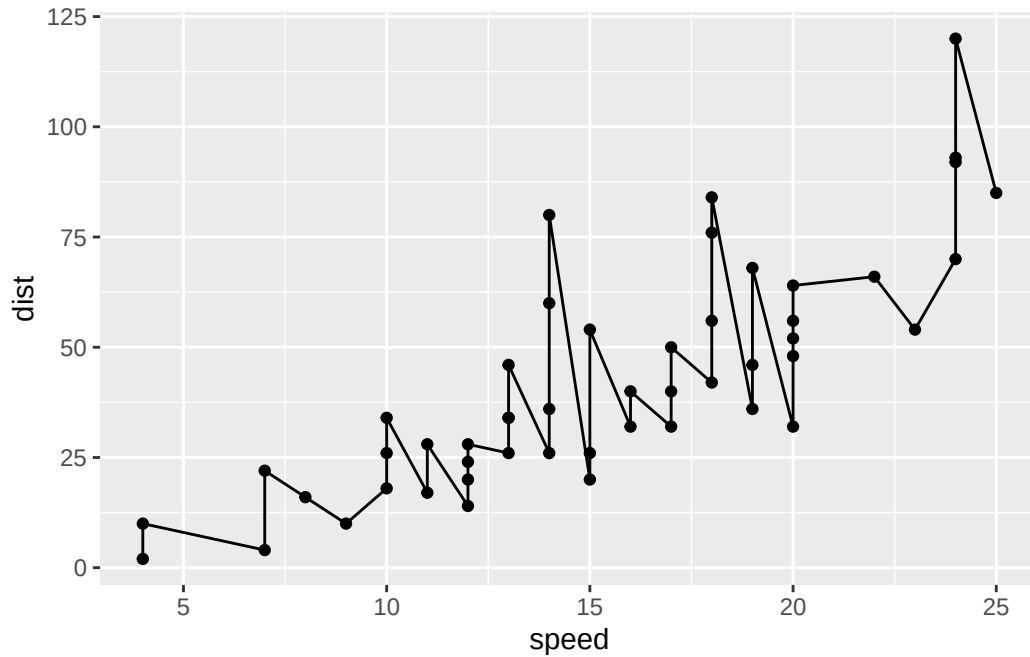
```
ggplot(cars)+  
  aes(x=speed, y=dist)+  
  geom_point()
```



Add some custom features

Lets add a trend line that shows the relationship between speed and distance.

```
ggplot(cars)+  
  aes(x=speed, y=dist)+  
  geom_point() +  
  geom_line()
```

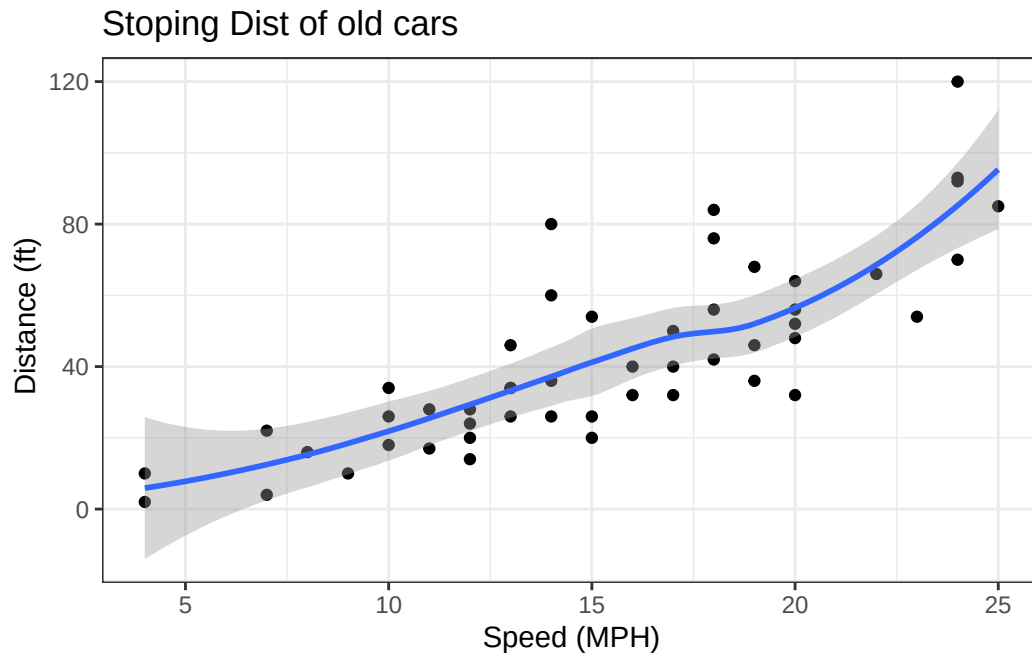


```
ggplot(cars)+
  aes(x=speed, y=dist)+
  geom_point() +
  geom_smooth()+
  theme_bw()+
  labs(title="Stopping Dist of old cars",
       x="Speed (MPH)",
       y=" Distance (ft)")
```

Ignoring unknown labels:

* titles : "Stopping Dist of old cars"

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'

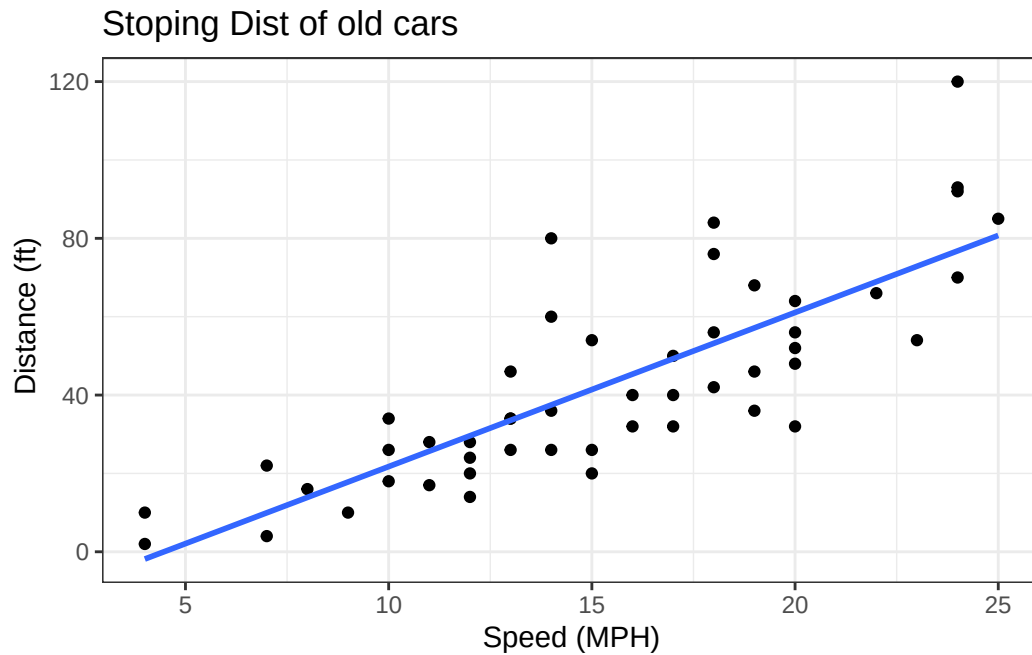


Q. Can you make the `geom_smooth()` function produce a linear

```
ggplot(cars)+
  aes(x=speed, y=dist)+
  geom_point() +
  labs(title="Stopping Dist of old cars",
        x="Speed (MPH)",
        y= " Distance (ft)")+
  geom_smooth(method="lm", se=FALSE)+
  theme_bw()
```

Ignoring unknown labels:

```
* titles : "Stopping Dist of old cars"
`geom_smooth()` using formula = 'y ~ x'
```



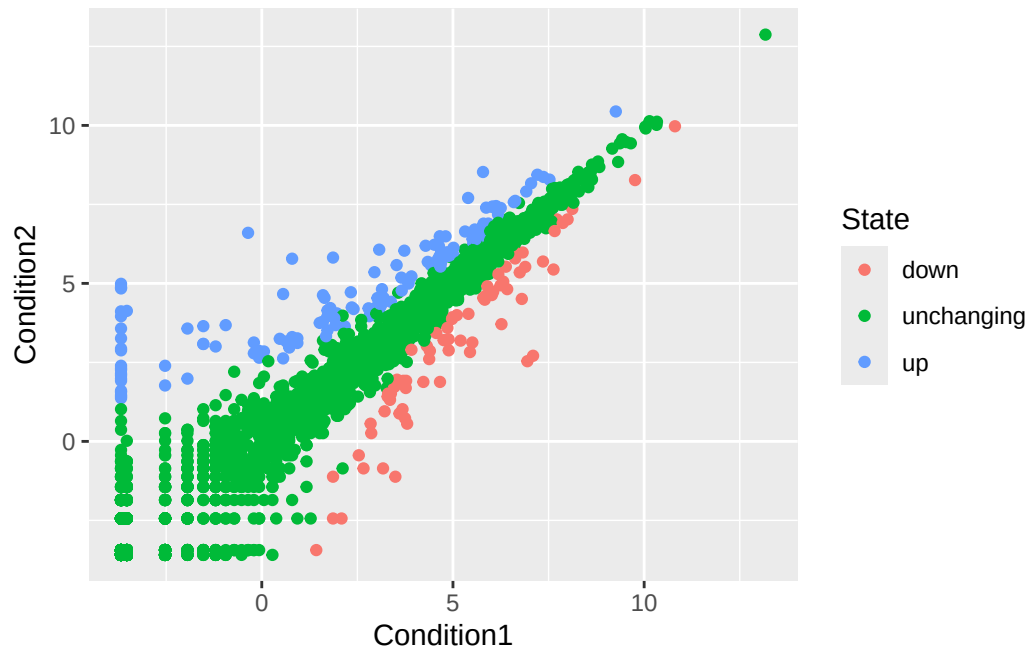
Gene expression Figure

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

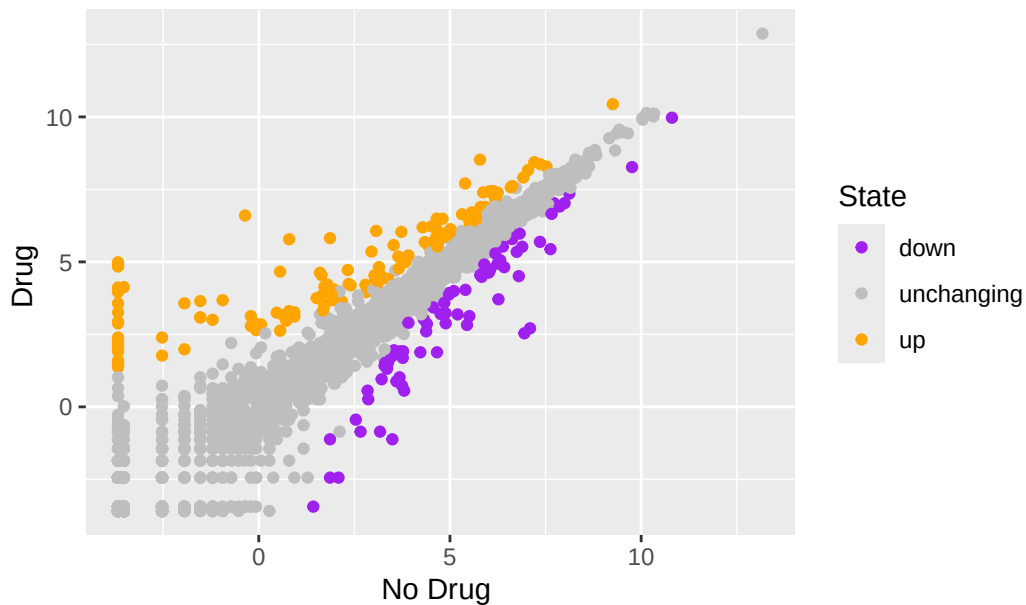
first plot attempt

```
ggplot(genes)+
  aes(x=Condition1, y=Condition2, col=State)+
  geom_point()
```



```
ggplot(genes)+
  aes(x=Condition1, y=Condition2, col=State)+
  geom_point()+
  scale_colour_manual(values=c("purple","grey","orange"))+ labs(x="No Drug ",
    y="Drug ",
    title="Gene Expression upon GLP-1 inhibitor treatment")
```

Gene Expression upon GLP-1 inhibitor treatment



Expression Chnages

```
#File location
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"

gapminder <- read.delim(url)
head(gapminder)
```

	country	continent	year	lifeExp	pop	gdpPercap
1	Afghanistan	Asia	1952	28.801	8425333	779.4453
2	Afghanistan	Asia	1957	30.332	9240934	820.8530
3	Afghanistan	Asia	1962	31.997	10267083	853.1007
4	Afghanistan	Asia	1967	34.020	11537966	836.1971
5	Afghanistan	Asia	1972	36.088	13079460	739.9811
6	Afghanistan	Asia	1977	38.438	14880372	786.1134

Q. How many entries (i.e.rows) are in this dataset?

```
nrow(gapminder)
```

```
[1] 1704
```

Q. How many “country” are in this dataset?

```
length(table(gapminder$country))
```

```
[1] 142
```

```
length(unique(gapminder$country))
```

```
[1] 142
```

let's make our first plot of the entire dataset:

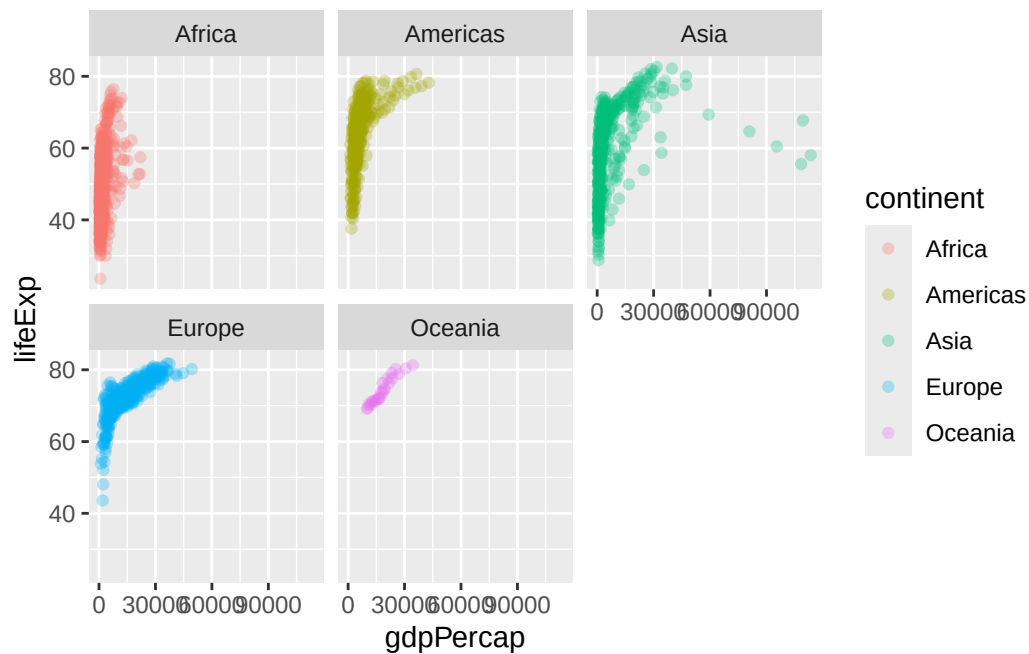
plot of “gdpPercap” vs. “lifeExp” colored by “continent”

```
p<- ggplot(gapminder)+  
  aes(gdpPercap,lifeExp,col=continent )+  
  geom_point(alpha=0.3)  
p
```



I can add more layers to ‘p’

```
p+
  facet_wrap(~continent)
```



Make a plot for 1977 and 2007

```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"
gapminder <- read.delim(url)
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

```
filter, lag
```

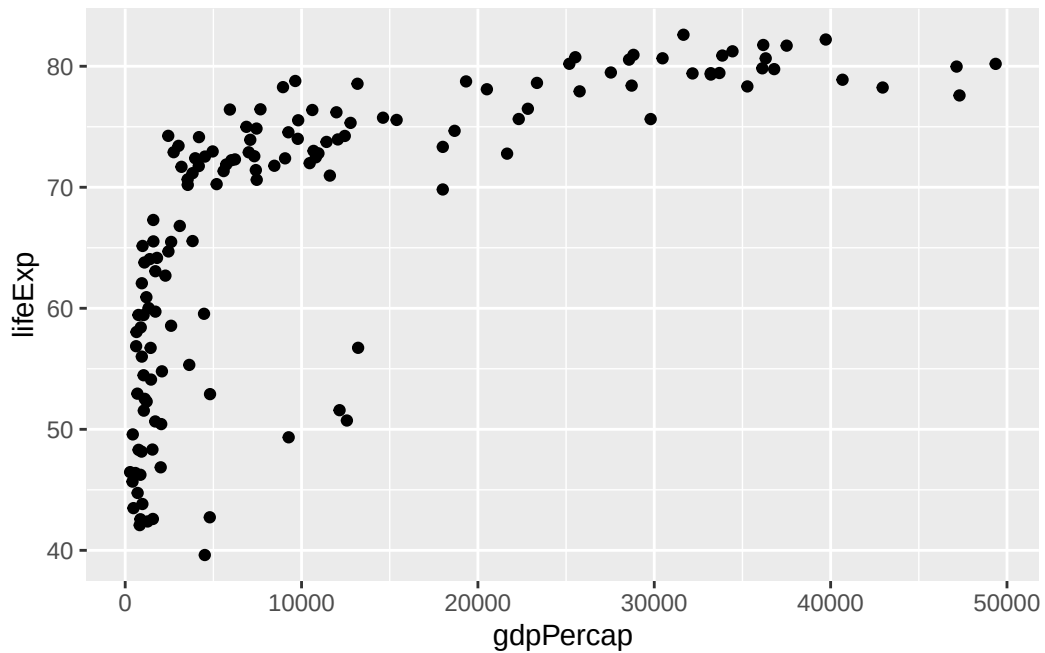
The following objects are masked from 'package:base':

```
intersect, setdiff, setequal, union
```

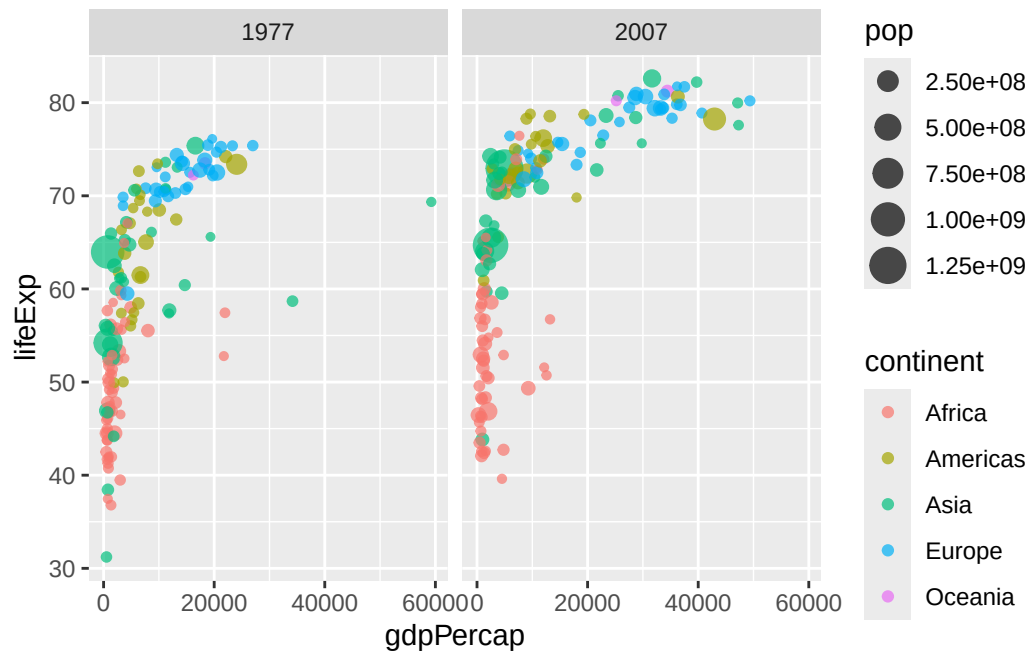
```
gapminder_both<-gapminder%>% filter(year==1977|year==2007)
```

```
gapminder_2007<-gapminder%>% filter(year==2007)
```

```
ggplot(gapminder_2007)+  
  aes(gdpPerCap,lifeExp)+  
  geom_point()
```

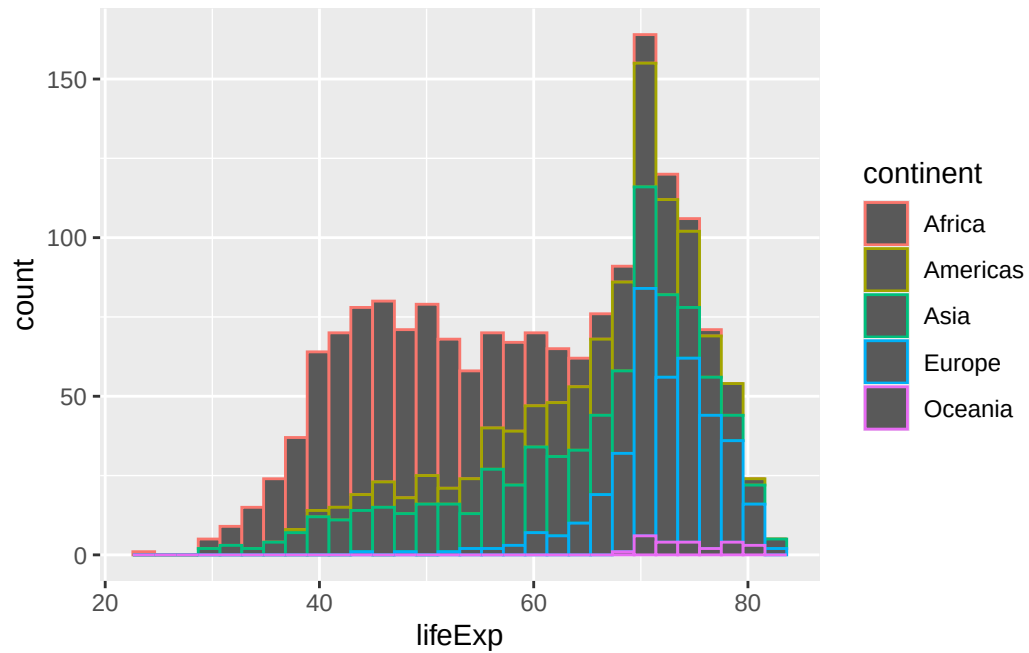


```
ggplot(gapminder_both)+  
  geom_point(aes(x=gdpPerCap, y=lifeExp, col=continent, size=pop),alpha=0.7)+  
  facet_wrap(~year)
```



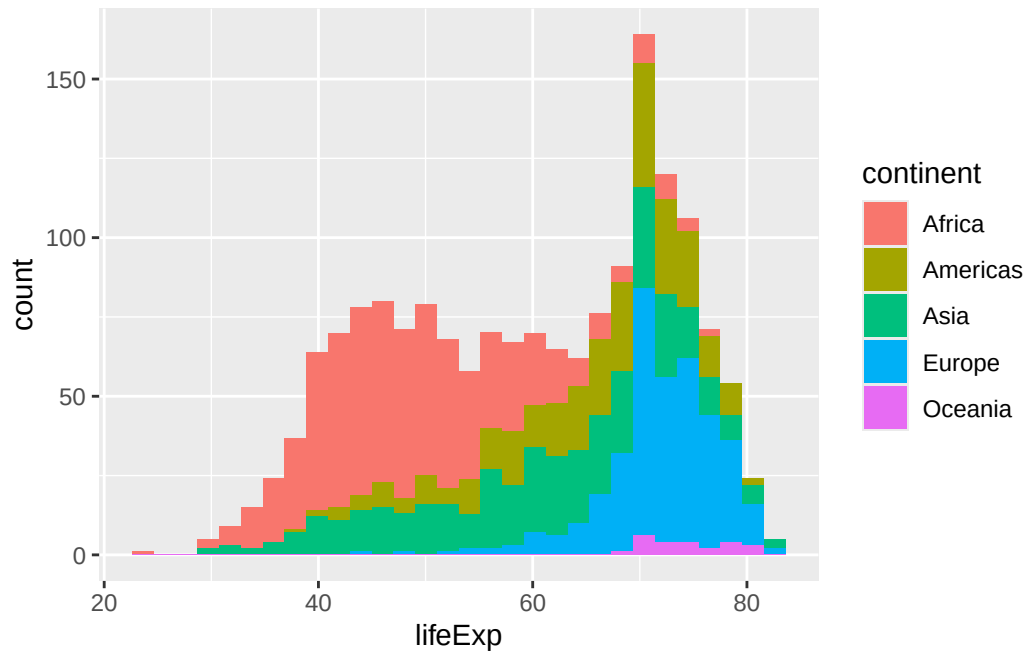
```
ggplot(gapminder)+
  aes(lifeExp,col=continent)+
  geom_histogram()
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



```
ggplot(gapminder)+  
  aes(lifeExp,fill=continent)+  
  geom_histogram()
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Q. Make a histogram of lifeExp facted by continent

```
ggplot(gapminder)+  
  aes(lifeExp)+  
  geom_histogram()+  
  facet_wrap(~continent)
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.

